Clase 5
Introducción a las Redes de Computadoras

Objetivos de la Clase

- Make a quick *Introduction*
- Introduce the concept of *Network Layers*
- Review key concepts of the *the TCP/IP Stack*
- Discuss *Key Concepts and Protocols* to understand how all fits together
- Talk about different aspects of *Performance and Reliability*
- Examine some *Security Concepts* vital when deploying applications

Agenda

1. **Introduction**
2. Network Layers
3. The TCP/IP Stack
4. Key Concepts and Protocols
5. Performance and Reliability
6. Security Concepts

What is a computer network?

“A computer network is a system that enables computers to **exchange data efficiently and reliably**.”

Why is this important for AI?

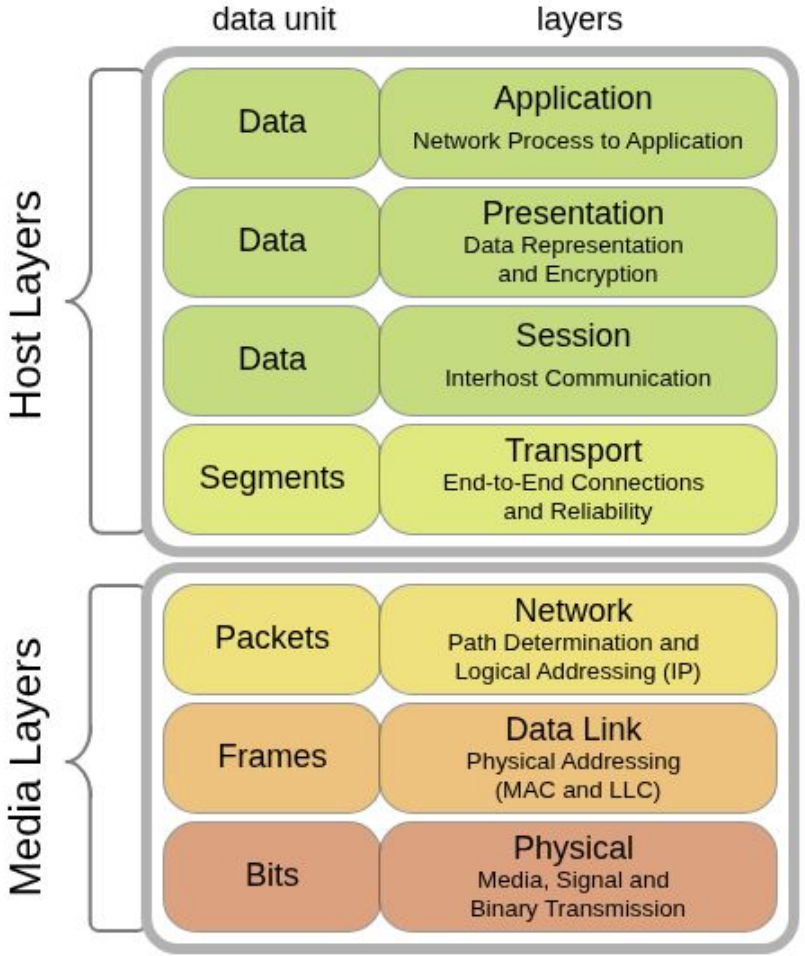
- Motivation: **AI models are never alone**, data pipelines, cloud inference, model serving, distributed training, IoT edge models.
- Examples:
  - ChatGPT-like models **rely on HTTP and APIs**.
  - **Distributed training** over clusters → high-speed interconnects (Ethernet, InfiniBand).
  - **Federated learning** → secure communication between clients.

Agenda

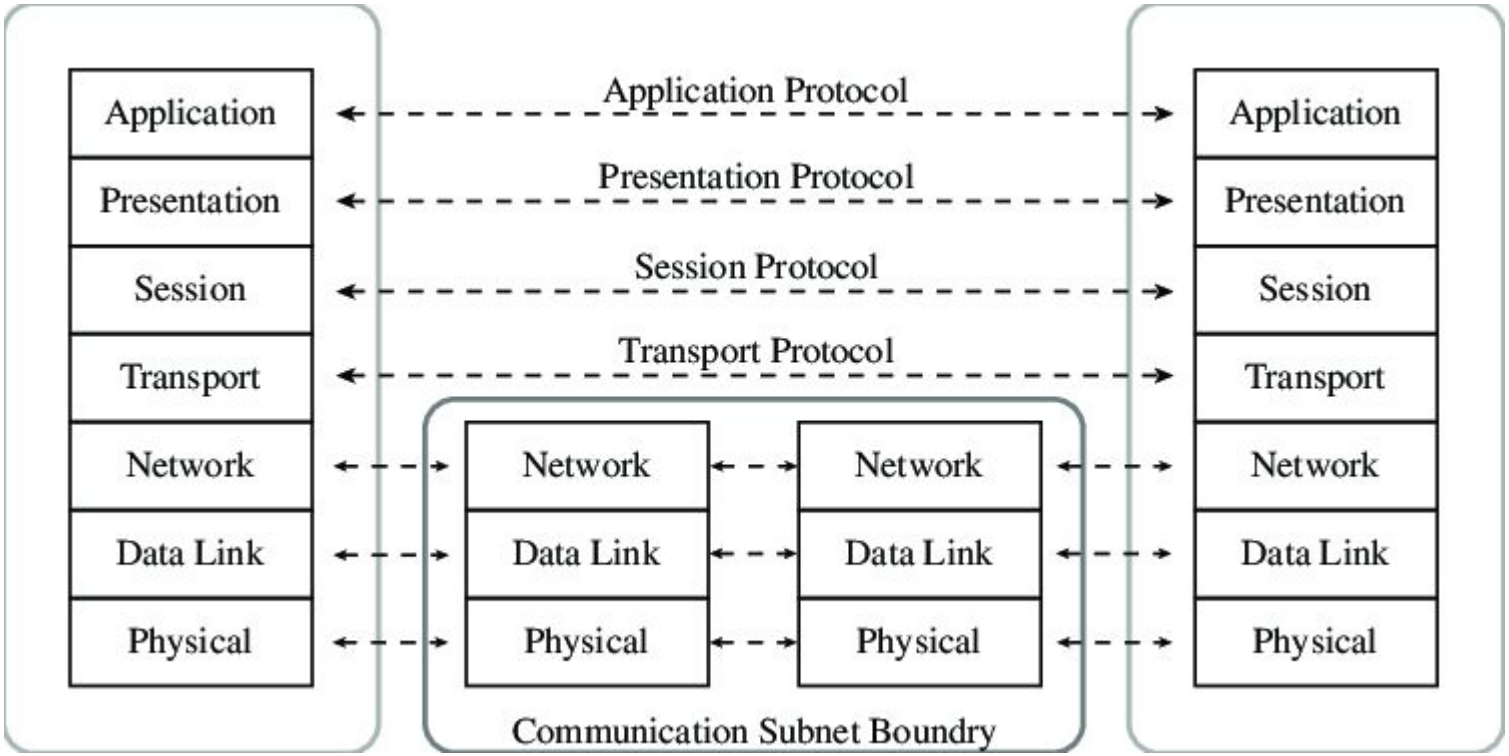
1. Introduction
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ISO/OSI Model

- ISO/OSI is a **framework** for understanding network communication in 7 abstract layers.
- Some layers work inside the host and others depend on the specific communication media.



The protocol stack

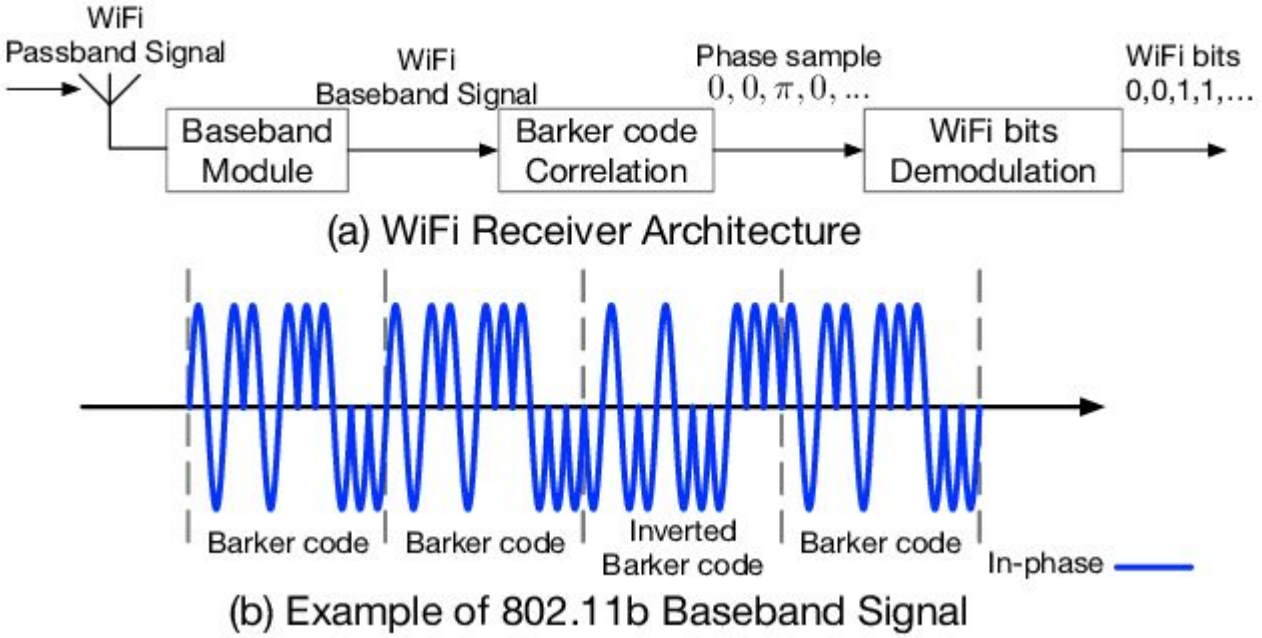


Ex: Ethernet





Ex: WiFi aka 802.11



Ex: IP over Avian Carriers



rfc-editor.org/rfc/rfc1149

[RFC Home] [TEXT|PDF|HTML] [Tracker] [IPR] [Errata] [Info page]

EXPERIMENTAL
Updated by: 2549, 6214 Errata Exist
Network Working Group D. Waitzman
Request for Comments: 1149 BBN STC
1 April 1990

A Standard for the Transmission of IP Datagrams on Avian Carriers

Status of this Memo

This memo describes an experimental method for the encapsulation of IP datagrams in avian carriers. This specification is primarily useful in Metropolitan Area Networks. This is an experimental, not recommended standard. Distribution of this memo is unlimited.

Overview and Rational

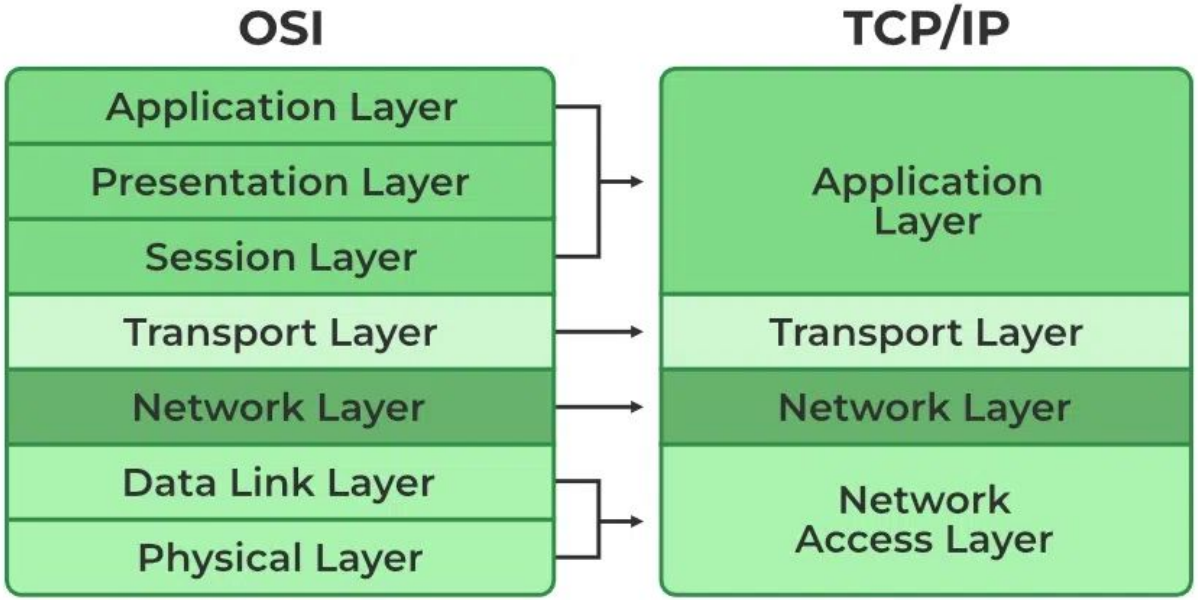
Avian carriers can provide high delay, low throughput, and low altitude service. The connection topology is limited to a single point-to-point path for each carrier, used with standard carriers, but many carriers can be used without significant interference with each other, outside of early spring. This is because of the 3D ether space available to the carriers, in contrast to the 1D ether used by IEEE802.3. The carriers have an intrinsic collision avoidance system, which increases availability. Unlike some network technologies, such as packet radio, communication is not limited to line-of-sight distance. Connection oriented service is available in some cities, usually based upon a central hub topology.

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3. **The TCP/IP Stack**
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The TCP/IP Stack

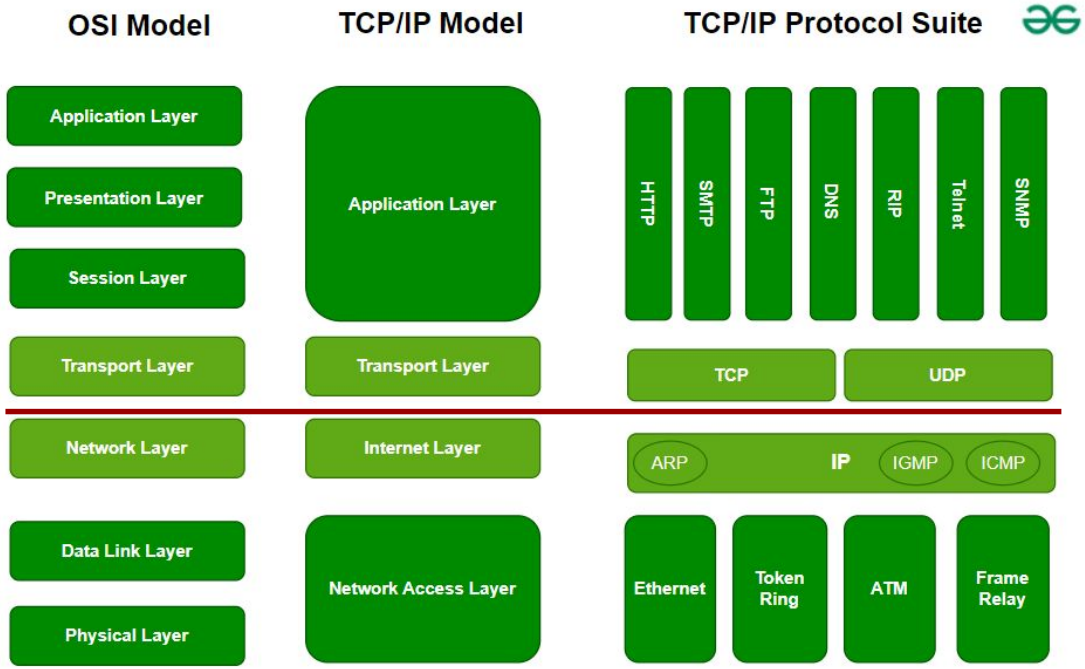
TCP/IP stack is a **suite of protocols** that maps to the OSI model for network communication.



The TCP/IP Stack

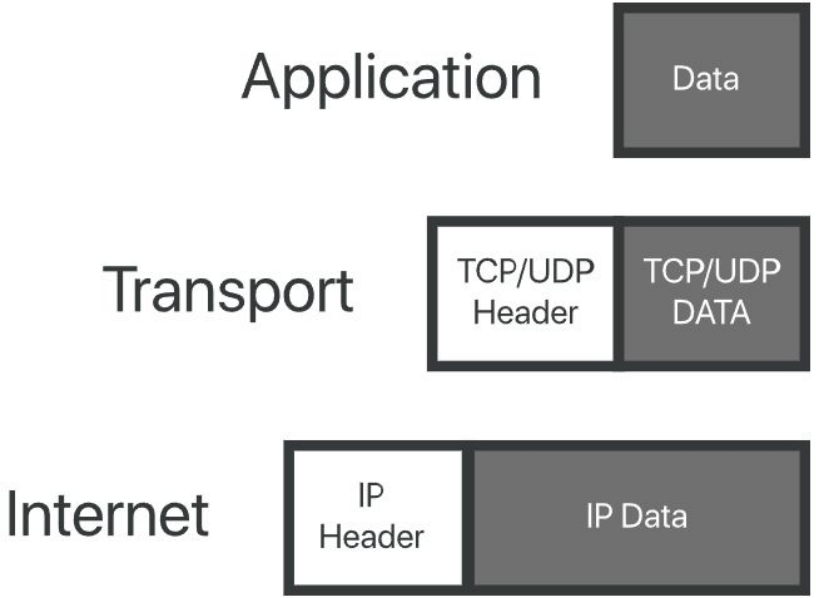
TCP/IP stack is a **suite of protocols** that maps to the OSI model for network communication.

- An **IP address identifies a device** on the network (e.g., 192.168.1.1).
- A **port number identifies a service on that device** (e.g., web server = 80).
- Together:
  - IP address + port number = **Unique endpoint** for a service.
- Example:
  - IP: 192.168.1.5
  - Port: 80 → HTTP service
  - Port: 443 → HTTPS service



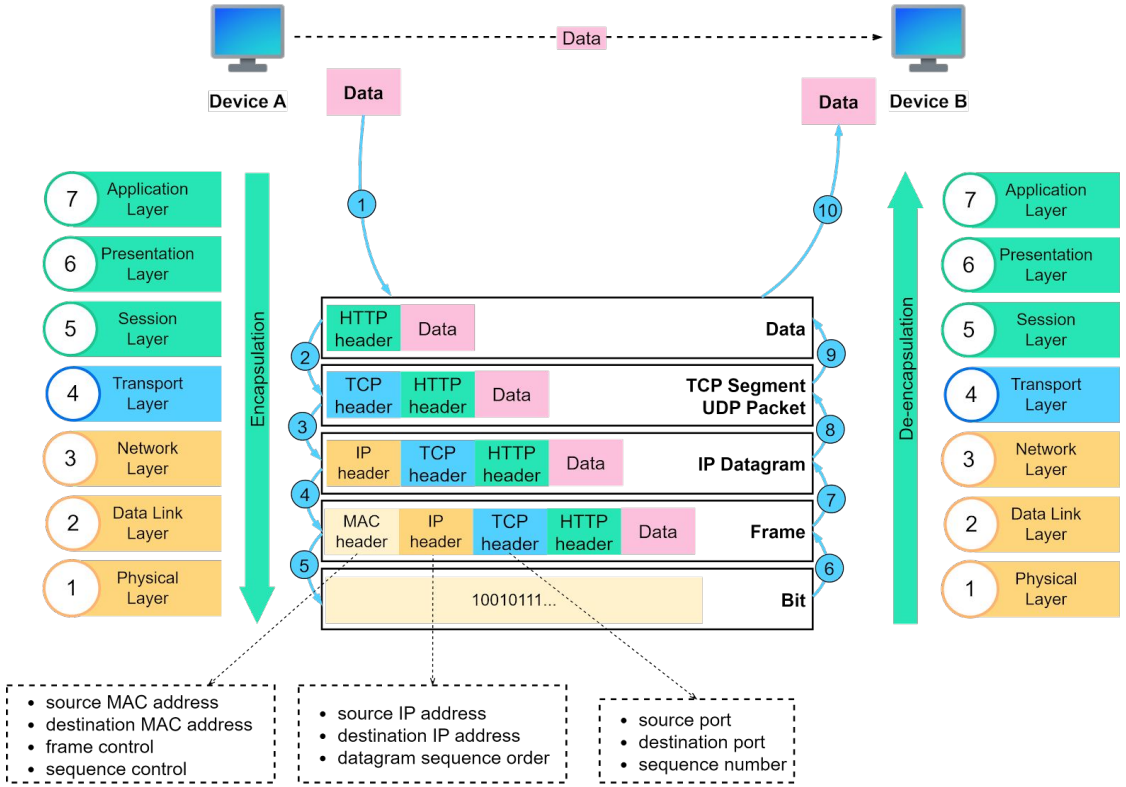
The TCP/IP Stack

Each layer **encapsulates** data of the previous layer.



The TCP/IP Stack

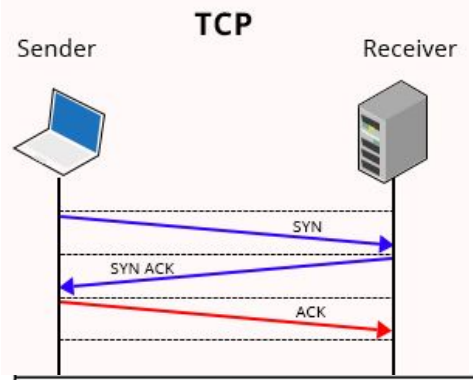
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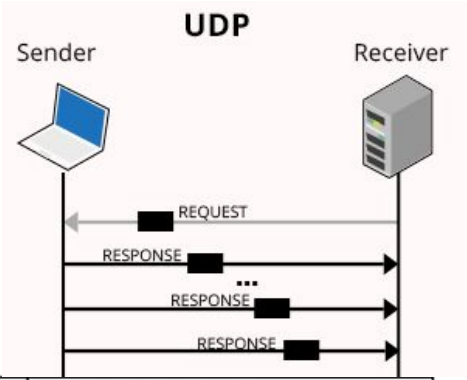


TCP vs UDP

TCP: Transmission Control Protocol

UDP: User Datagram Protocol





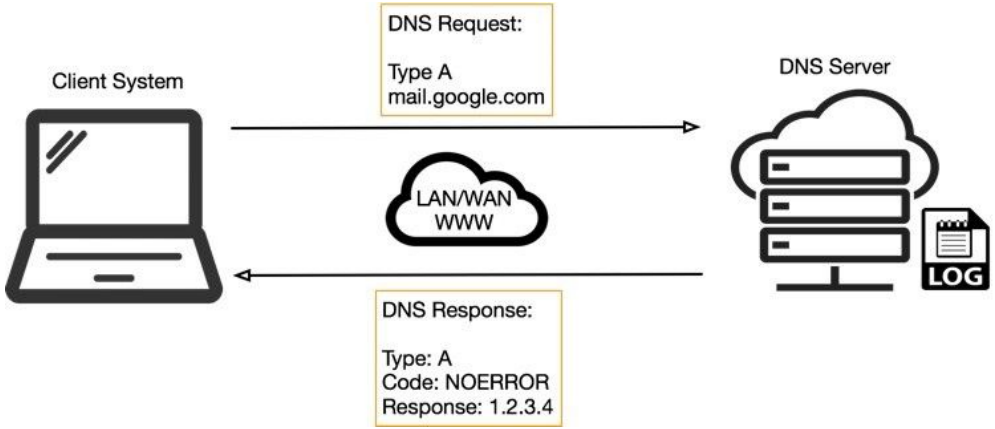
connection-oriented	connectionless
Delivery guarantee	No Delivery guarantee
Acknowledgement segments	No Acknowledgement
Slow Service	Fast Service
Sequencing	No Sequencing
Resend segment after drop	No Resend
Window Flow Control	No Flow Control
Error Checking	No Error Checking

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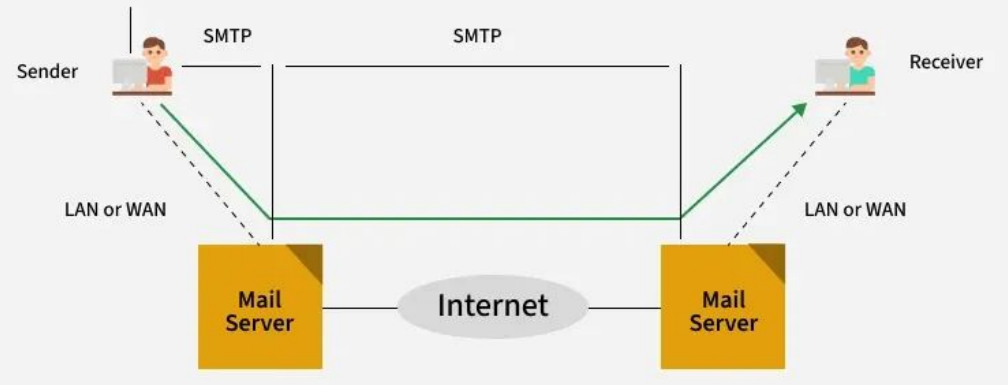
DNS

- **DNS** (Domain Name Service) is a protocol that resolves **human-readable domain names to IP addresses**.
- Port UDP/53



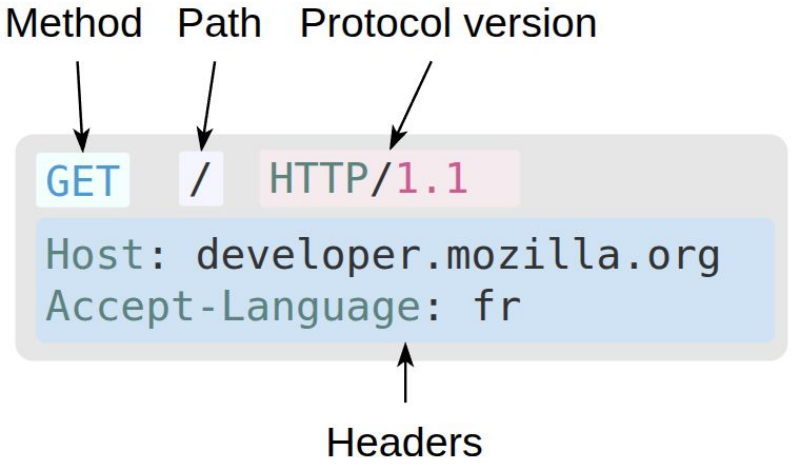
SMTP

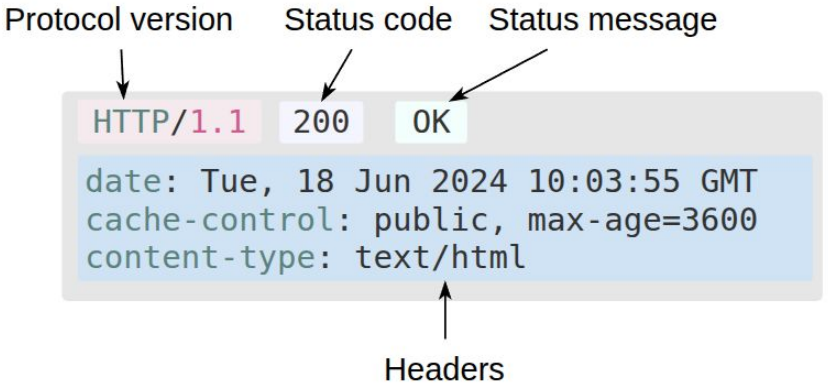
- **SMTP** (Simple Mail Transfer protocol) is a protocol used to send emails.
- Port TCP/25



HTTP

- Used for transferring **web content** over the internet.
- Request-response cycle
  - **Client sends request** to the server
  - **Server processes request and send a response** to the client
- Port TCP/80 or TCP/443 for HTTPS

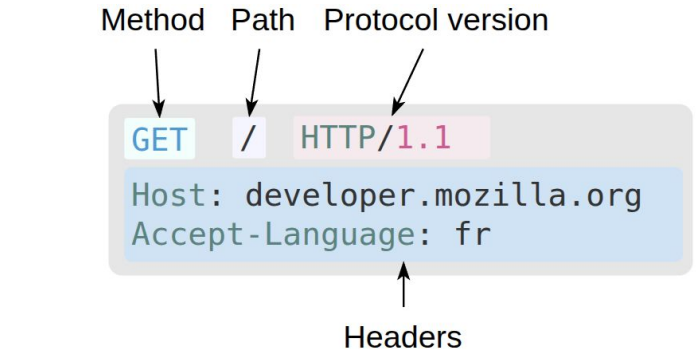


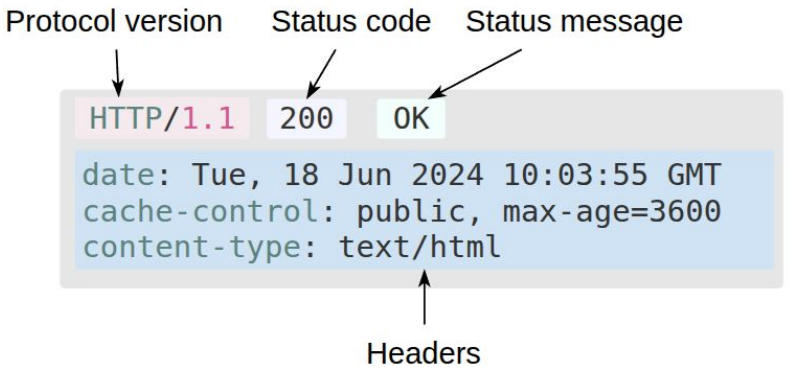


HTTP

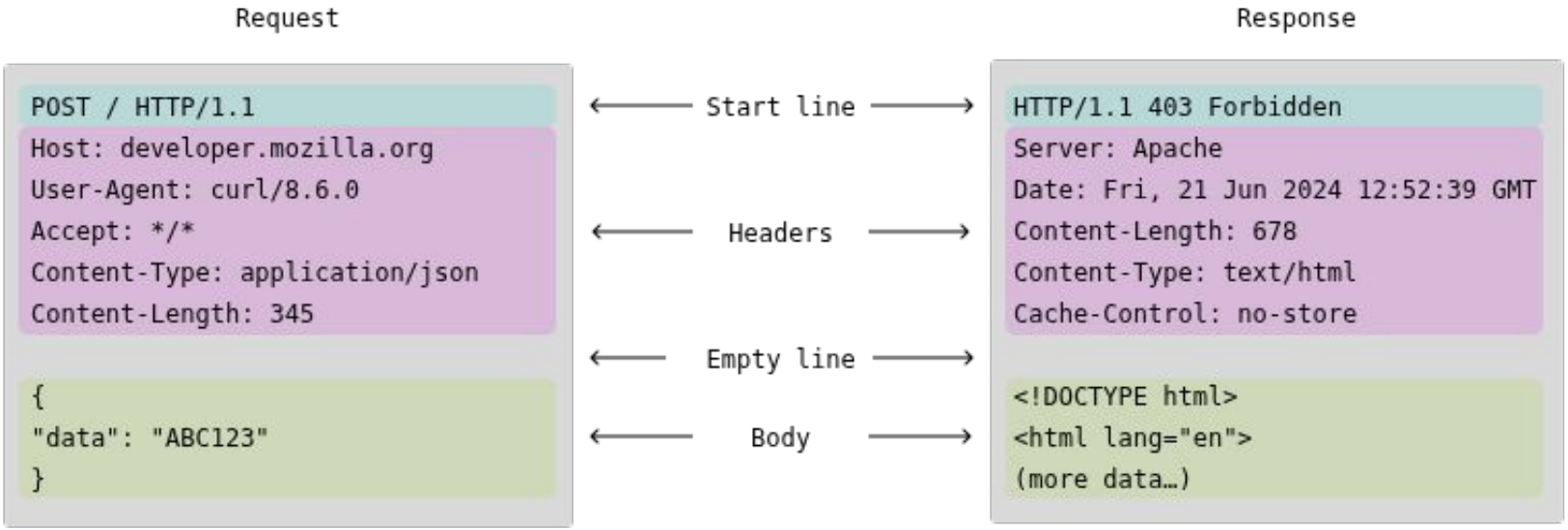
- HTTP **Request** Structure:
  - **Method**: Describes the action (e.g., GET, POST, PUT, DELETE).
  - **URL**: Specifies the resource (e.g., https://example.com/index.html).
  - **Headers**: Provide additional information (e.g., User-Agent, Accept).
  - **Body**: Contains data being sent (e.g., form data in POST).

- HTTP **Response** Structure:
  - **Status Code**: Indicates the result of the request (e.g., 200 OK, 404 Not Found).
  - **Headers**: Provide metadata (e.g., Content-Type, Date).
  - **Body**: Contains the actual data being returned (e.g., HTML content, JSON response).





HTTP


Request
Response
POST / HTTP/1.1
Host: developer.mozilla.org
User-Agent: curl/8.6.0
Accept: */*
Content-Type: application/json
Content-Length: 345
{
"data": "ABC123"
}
Start line
Headers
Empty line
Body
HTTP/1.1 403 Forbidden
Server: Apache
Date: Fri, 21 Jun 2024 12:52:39 GMT
Content-Length: 678
Content-Type: text/html
Cache-Control: no-store
<!DOCTYPE html>
<html lang="en">
(more data…)

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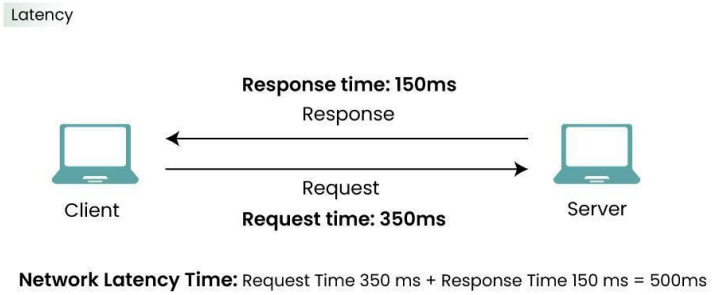
Key Concepts

- **Performance**: How well a network handles data transfer, typically measured in terms of **speed and efficiency**.

- **Reliability**: The ability of a network to consistently **deliver data correctly, without loss or errors**.

Metrics

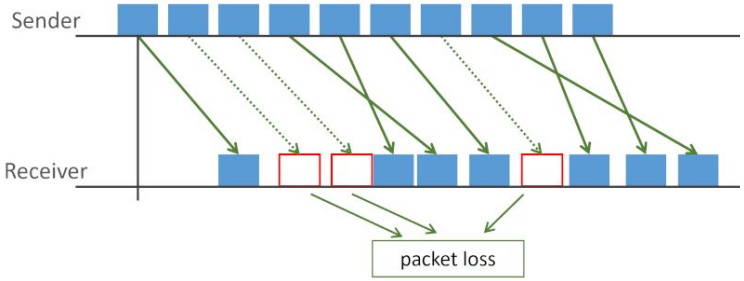
- Latency
  - The **time it takes for a packet of data to travel** from the sender to the receiver.



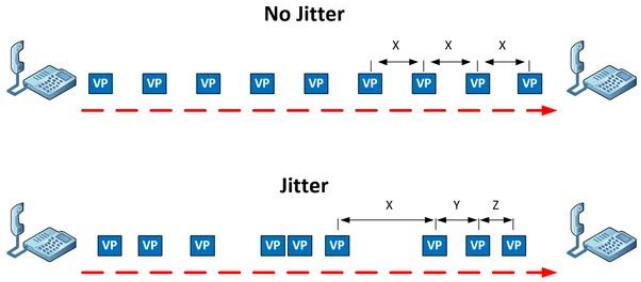
- Bandwidth (Throughput)
  - The **maximum rate at which data can be transmitted** over a network, often measured in bits per second (bps).

Metrics

- Packet Loss
  - Occurs when **data packets fail to reach** their destination.



- Jitter
  - The **variation in packet arrival times**, often caused by network congestion or route instability.



Trade-offs

- **Low Latency vs High Throughput**: Some use case require low latency over high throughput, or vice versa.

- **Reliability vs Speed**: Reliable protocols like TCP introduce delays due to error checking and retransmissions, while protocols like UDP prioritize speed.

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6. **Security Considerations**

Security Considerations

- Communications networks shouldn’t be assumed to be secure by default.
- Network security tryad:
  - **Integrity**: Ensuring data is accurate and unaltered during storage or transmission.
  - **Confidentiality**: Protecting data from unauthorized access or disclosure.
  - **Availability**: Ensuring data and services are accessible when needed.

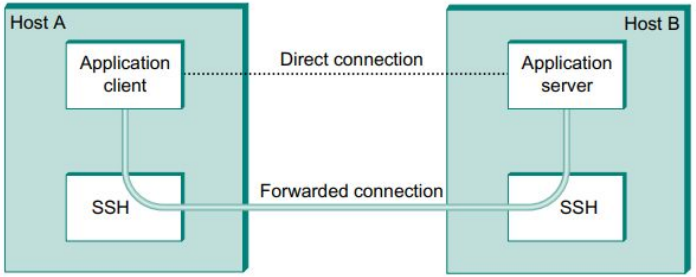


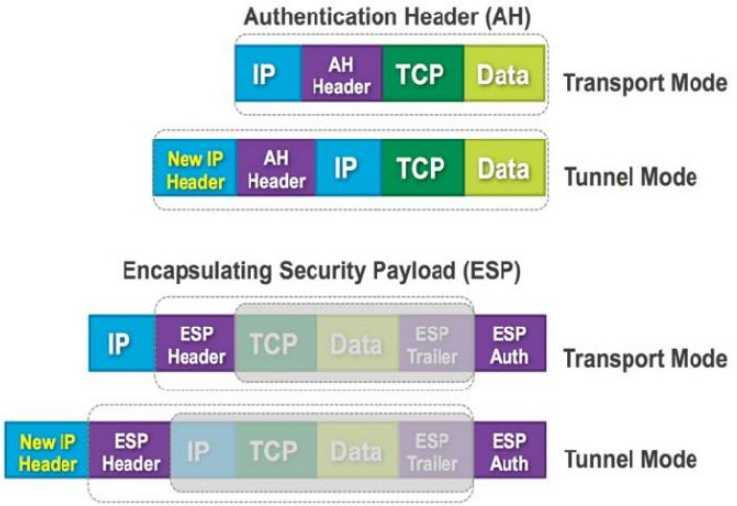
Common of Attacks

- Unauthorized access:
  - An attacker accesses a restricted resource either due to ineffective authentication or impersonation.
- Denial of Service (DoS):
  - Attackers flood the network with traffic, overwhelming resources and causing service outages.
- Man-in-the-Middle (MITM):
  - Attackers intercept and potentially alter communications between two parties.
- Packet Sniffing:
  - Attackers capture data packets to extract sensitive information such as passwords.
- IP Spoofing:
  - Attackers impersonate trusted devices, often to gain unauthorized access or cause disruption.

Encryption and VPNs

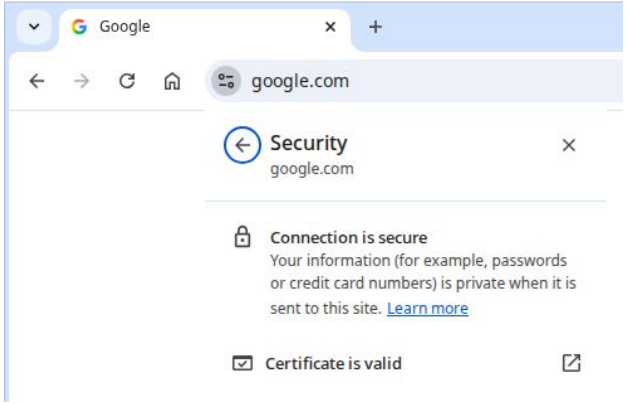
- Data Encryption **ensures data is unreadable during transit**
  - SSL/TLS: Common for securing web communications (HTTPS).
  - SSH: Secure protocol to log into remote systems.
  - IPsec: Encrypts data at the IP layer, securing communication between devices.
- **Virtual Private Networks (VPNs)**
  - VPNs create a secure, encrypted tunnel for data over an untrusted network (like the internet).
  - Protect against eavesdropping and secure remote access to private networks.

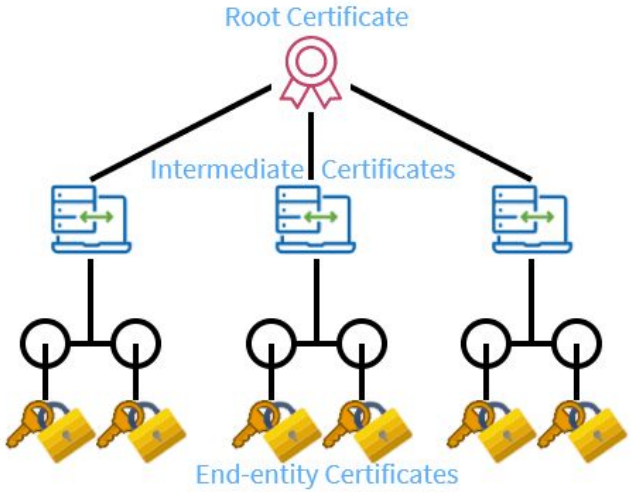




SSL/TLS Certificates

- Ensure that the data exchanged between the client and server is confidential, integrity-protected, and authenticated.
- **Certificate Authorities (CAs) are trusted organizations that issue SSL/TLS certificates** to websites after verifying their identity.
- The **CA signs the certificate** to ensure that the public key in the certificate belongs to the organization that owns the domain.





Dudas?